Duong Ta

Applied Economist & Data Analyst | Ph.D. Candidate, Agricultural and Applied Economics, Virginia Tech
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Professional Summary

Applied economist with 15+ years of experience (2010–present) researching agriculture, climate, and trade in Vietnam and the U.S. Skilled in causal inference (difference-in-differences, instrumental variables, double/debiased machine learning) and large-scale survey data analysis using Python, Stata, and R. Author or co-author of 8 journal publications, with current research on climate stress and milk production and a silvopasture manuscript under review.

Technical Skills

- **Programming & Statistical Software:** Stata, Python, R, SQL, LaTeX
- **Econometric & Causal Inference Methods:** Difference-in-Differences, Panel Data Models, Double/Debiased Machine Learning, Cost-Benefit Analysis
- **Machine Learning & Data Science:** Scikit-learn, Pytorch, TensorFlow
- **Web Data Collection:** Python-based scraping and AI-assisted crawling tools
- **Geospatial & Remote Sensing:** Google Earth Engine, QGIS
- **Reproducibility & Collaboration Tools:** Git, GitHub, Quarto

Professional Experience

- **Virginia Tech [🌐]** *2023 – present*
  Graduate Research & Teaching Assistant USA
  - Develop the current job market paper, "Estimating impacts of climate stress on US milk production: a panel data with debiased machine learning approach", as first author.
  - First author of "Economic Feasibility and Carbon Sequestration of Silvopasture in the Southeastern U.S." Submitted to a journal and under review. Presented at the 2025 AAEA-WAEA Annual Meeting.
  - Support instruction for undergraduate and graduate courses in the Department of Agricultural and Applied Economics, including lab sections and grading.
- **The International Center for Tropical Agriculture (CIAT), Asia Hub [🌐]** *2018 – 2022*
  Research Assistant / Data Analyst (held concurrently with the VIE research post) Vietnam
  - Built questionnaires and CSPro data-entry applications for two surveys of about 1,000 farm households each (Mekong Delta rice under salinity, 2019–2022; cassava value chains, 2019–2021). As field supervisor, trained teams of about 12 enumerators and supervised data collection.
  - Managed and cleaned the resulting survey data and built econometric analysis pipelines in R and Stata, contributing to two peer-reviewed publications (*Climate and Development*, 2022; *Journal of the Agricultural and Applied Economics Association*, 2025).
- **Vietnam Institute of Economics (VIE) [🌐]** *2010 – 2023*
  Researcher Vietnam
  - Led projects on farm and non-farm employment in Vietnam's rural economy (2022) and agricultural structure under climate change in the Mekong Delta (2021), including proposals, data analysis, and final research reports.
  - Studied trade and structural adjustment, non-farm livelihoods, agricultural income diversification, and climate adaptation among Vietnamese households.
  - Authored or co-authored 7 journal publications (2015–2023) on rural livelihoods, climate adaptation, and structural transformation.

Selected Publications

[1] Mills et al. (2025), incl. Ta. "Salinity Inundation, Profitability, and Rice Farming Exits in Mekong Delta of Vietnam". *J. of the Agricultural and Applied Economics Assoc.*, 4(2), 170-186.

[2] Mills et al. (2022), incl. Ta. "Intensive and Extensive Rice Farm Adaptations in Salinity-Prone Areas of the Mekong Delta". *Climate and Development*, 15(2), 162-176.

[3] Nguyen Dinh Chuc and Ta Phuc Duong (2019). "Trade, Structural Adjustments and Productivity Growth in Vietnam: The Shift to Services". *J. of Southeast Asian Economies*, 36(2), 256-273.

[4] Ta Phuc Duong (2015). "Effects of Capital Flows on Housing Prices in Emerging Economies: A Panel Approach". *Vietnam's Socio-Economic Development*, No. 81 (April 2015).

Plus 4 Vietnamese-language publications and 2 current papers, including the job market paper, listed in the academic CV.

Education

- **Virginia Tech** *2023 – Expected Summer 2027*
  Ph.D. in Agricultural and Applied Economics USA
- **University of Greenwich** *2014 - 2015*
  MSc in Business and Financial Economics United Kingdom
- **The National Economics University** *2004 - 2008*
  Bachelor in Economics Vietnam

Certifications & Additional Information

Certifications: Machine Learning Specialization, Stanford University via Coursera (3 courses), 2025

Affiliations: Agricultural and Applied Economics Association (AAEA)

Languages: Vietnamese (native), English (fluent)